

Algorithmic Trading System

How to Run

Practical setup and execution guide

9. How to Run

This section provides the practical instructions required to run the algorithmic trading system locally. The application is Dockerised and can be run locally without installing the Python dependencies directly on the host system.

9.1 Requirements

- Python 3.11 or later.
- Docker Desktop with Docker Compose.
- Git.
- Access to the private Google Sheet used for application configuration.
- Google Cloud service-account credentials for reading the configuration sheet.
- Google OAuth credentials for authenticated Streamlit access.
- The Meta Llama 3.1 8B Instruct GGUF model used by the application.

9.2 Project Setup

Clone or copy the project repository to the local machine. The project directory should contain the application source code, Docker configuration, and the following local resources:

- `credentials/apziva-config-reader.json`
- `.streamlit/secrets.toml`
- `models/Meta-Llama-3.1-8B-Instruct-Q4_K_M.gguf`

The credential file, Streamlit secrets file, and model file contain local resources and should not be committed to version control.

9.3 Configuration

Application configuration is stored in the private Google Sheet and is refreshed periodically by the application. A local JSON cache provides a fallback when the Google Sheet cannot be reached.

Sensitive credentials are kept separately from the application configuration. The Streamlit secrets file contains the credentials and configuration required for authenticated access and the Google Sheet integration.

9.4 Build and Run

From the project directory, run:

```
docker compose build
```

```
docker compose up
```

Docker builds the application image and starts the Streamlit application. The quantised Llama model is mounted into the container at runtime rather than being included in the Docker image.

9.5 Accessing the Application

Once the container has started, open the local Streamlit application in a web browser at:

```
http://localhost:8502
```

If authentication is enabled, sign in using the configured Google OAuth provider before using the application.

9.6 Stopping the Application

To stop the running containers, use:

```
docker compose down
```

9.7 Important Notes

- The system operates using a simulated portfolio and read-only market access. It does not place real cryptocurrency trades.
- The Binance integration is read-only and does not require trading credentials.
- The Google service-account credentials and Streamlit secrets must remain outside version control.
- The Llama GGUF model is excluded from the Docker build context and mounted into the container at runtime.
- For a remote deployment, the Google OAuth redirect URI must be updated to match the deployed HTTPS address.

End of practical run guide